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Artificial Intelligence for Clinical Prediction: Exploring Key Domains and Essential Functions

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ABSTRACT

Background: Clinical prediction is integral to modern healthcare, leveraging current and historical medical data to forecast health outcomes. The integration of Artificial Intelligence (AI) in this field significantly enhances diagnostic accuracy, treatment planning, disease prevention, and personalised care leading to better patient outcomes and healthcare efficiency.

Methods: This systematic review implemented a structured four-step methodology, including an extensive literature search in academic databases (PubMed, Embase, Google Scholar), applying specific inclusion and exclusion criteria, data extraction focusing on AI techniques and their applications in clinical prediction, and a thorough analysis of the collected information to understand AI's roles in enhancing clinical prediction.

Results: Through the analysis of 74 experimental studies, eight key domains, where AI significantly enhances clinical prediction, were identified: (1) Diagnosis and early detection of disease; (2) Prognosis of disease course and outcomes; (3) Risk assessment of future disease; (4) Treatment response for personalised medicine; (5) Disease progression; (6) Readmission risks; (7) Complication risks; and (8) Mortality prediction. Oncology and radiology come on top of the specialties benefiting from AI in clinical prediction.

Discussion: The review highlights AI's transformative impact across various clinical prediction domains, including its role in revolutionising diagnostics, improving prognosis accuracy, aiding in personalised medicine, and enhancing patient safety. AI-driven tools contribute significantly to the efficiency and effectiveness of healthcare delivery.

Conclusion and recommendations: AI's integration in clinical prediction marks a substantial advancement in healthcare. Recommendations include enhancing data quality and accessibility, promoting interdisciplinary collaboration, focusing on ethical AI practices, investing in AI education, expanding clinical trials, developing regulatory oversight, involving patients in the AI integration process, and continuous monitoring and improvement of AI systems.

Introduction

The field of clinical prediction is a cornerstone of modern healthcare. This process involves the use of current and historical medical data to forecast future health outcomes [1]. Clinical prediction helps in the early detection and prevention of diseases, enhancing the accuracy of diagnoses, and improving treatment planning. This results in better patient outcomes and improved efficiency of healthcare systems [2,3]. By processing extensive and complex medical data quickly and with high precision, Artificial intelligence (AI) algorithms can identify patterns and correlations that might be beyond the scope of human analysis. These algorithms are designed to continuously learn and improve from new data, which enhances their predictive accuracy over time [2,3]. By thoroughly analysing medical images, lab results, and patient histories, AI can identify signs of diseases such as cancer or cardiovascular disorders more accurately and at an earlier stage than traditional methods [4]. AI is instrumental in the advancement of personalised medicine. It helps doctors understand how different diseases progress in individual patients and predicts how they might respond to various treatments. This leads to more tailored treatment plans,

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maximising effectiveness while minimising potential side effects [5]. Furthermore, AI plays a vital role in risk assessment. It can identify patients who are at high risk of developing certain conditions based on genetic, lifestyle, and environmental factors. This enables the implementation of preventive measures delaying the onset of diseases [6].

AI also assists in predicting patient readmission risks and potential complications during medical and surgical procedures. This helps in planning and patient counselling, improving overall patient safety [7]. Moreover, AI's ability to predict mortality risks in critical and palliative care settings is invaluable. It helps in making crucial decisions regarding the intensity of treatment and end-of-life care, ensuring that patients receive appropriate and compassionate care tailored to their individual needs [8].Therefore, AI's integration into clinical prediction represents a significant leap forward in healthcare. Its ability to analyse and interpret vast amounts of medical data not only improves the accuracy of predictions but also personalizes patient care, enhances treatment effectiveness, and promotes preventive healthcare [9]. Accordingly, this systematic review seeks to extensively examine AI's role in enhancing clinical prediction functions. It focuses on identifying key areas, and clinical specialties, where AI enhances clinical prediction. This is essential for assessing AI's readiness for broader adoption, identifying research gaps, and guiding future research and development.

Methods

To conduct this systematic review, a structured four-step methodology was implemented to ensure a comprehensive and meticulous examination of relevant literature. The initial step included an extensive search across several academic databases including PubMed, Embase, and Google Scholar. The focus was on articles published in English from 2019 onwards, using keywords such as "artificial intelligence," "clinical prediction," "healthcare analytics," "predictive modelling," and "patient outcomes." This search aimed to gather peer-reviewed articles and primary studies that explored the use of AI in clinical prediction scenarios. The second step involved the development and application of specific inclusion and exclusion criteria. Studies were selected if they primarily investigated AI's role in enhancing clinical prediction, focusing on aspects like predictive accuracy, patient outcomes, and decision-making processes. Excluded were studies not centrally concerned with clinical prediction, those lacking empirical data, or with ambiguous methodologies. In the third step, relevant data were extracted from the selected studies, concentrating on the primary AI techniques used, significant findings, specific applications in clinical prediction, and the observed limitations and future recommendations. This data was then aggregated to highlight crucial areas where AI contributes to clinical prediction, identifying trends, barriers, and potential for further application in healthcare. The final step involved a thorough analysis of the collated information. This analysis aimed to clarify the roles of AI in enhancing

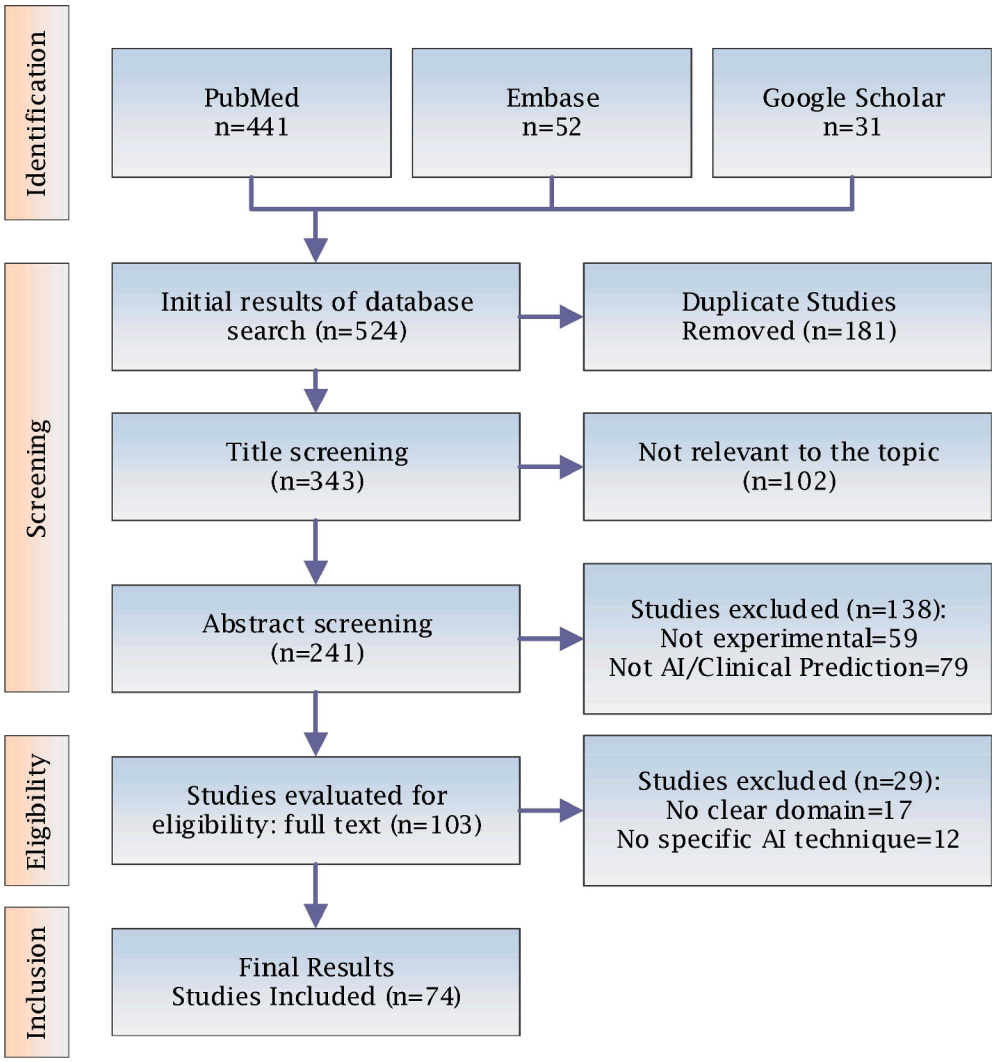

Fig. 1. PRISMA flowchart of study selection and inclusion process.

clinical prediction, noting the improvements in prediction accuracy, efficiency, and patient outcomes. Additionally, the analysis addressed the challenges in implementing AI in this field, including ethical implications, data security issues, and the integration with existing clinical workflows.

Results

Searching PubMed, Google Scholar, and Embase, 524 studies were found. After removing duplicates, 343 unique studies were identified. When inclusion and exclusion criteria were applied, 102 studies were excluded after title screening and another 138 were excluded after abstract screening. After full-text examination, 74 studies, out of 103, were included. Fig. 1 shows the study selection and inclusion processes.

Through careful qualitative analysis, this systematic review identified eight domains where AI has the potential to significantly enhance clinical prediction. The first domain, Diagnosis, involves a critical aspect of medical care; predicting the presence or absence of a disease or condition. This is done by analysing symptoms, clinical tests, and other patient data. Effective diagnostic predictions are crucial for early detection and timely treatment of diseases and medical conditions [9,10]. Among the included 74 studies, 24 discussed the role of AI-based clinical prediction for diagnosis. The second domain, Prognosis, is centred around predicting the likely course and outcomes of a disease or condition once diagnosed. This knowledge helps healthcare professionals in devising more effective treatment plans and providing patients with realistic expectations about their conditions [2,11]. This domain was discussed by 38 studies. In the third domain, Risk Assessment, the focus shifts to predicting the likelihood of a patient developing a disease or condition in the future. This prediction is based on various factors such as genetics, lifestyle, environmental exposures, and existing health conditions. Accurate risk assessment is pivotal in preventive medicine and health promotion, allowing for early interventions that can significantly alter a patient's health trajectory [12,13]. This domain was discussed by 14 studies. The fourth domain, Treatment Response, go deeper into the area of personalised medicine. Here, the aim is to predict how a patient will respond to a specific treatment or therapy. This is particularly important as it helps in tailoring treatments to individual patients, known also as personalised medicine, thereby enhancing the effectiveness of therapeutic interventions and reducing the likelihood of adverse reactions [14,15]. This domain was discussed by 22 studies.

Disease Progression, the fifth domain, is crucial for managing chronic diseases such as diabetes, heart disease, and neurological disorders. Predicting how a disease will progress over time assists healthcare professionals in planning long-term treatment strategies and anticipating future care needs, ultimately improving the quality of life for patients with chronic conditions [2,16]. This domain was discussed by nine studies. The sixth domain, Readmission Risks, utilizes predictive models to identify patients who are at high risk of being readmitted to the hospital after discharge. This knowledge enables healthcare providers to offer targeted interventions, aiming to reduce readmission rates, which is a key indicator of quality healthcare [17,18]. This domain was discussed by three studies. In the seventh domain, Complication Risks, the focus is on predicting the risk of complications, both during and after medical procedures or treatments [19,20]. This domain was discussed by nine studies. Finally, the eighth domain, Mortality Prediction, is particularly relevant in critical care and palliative care settings. Predicting the risk of mortality is important for decision-making regarding the intensity of treatment and end-of-life care planning [21,22]. This domain was discussed by 20 studies.

Table 1 and Fig. 2 show the eight domains where AI supports clinical prediction. Table 2 shows mapping of the 74 studies to the identified eight domains of AI-based clinical prediction. Accordingly, Fig. 3 shows the potential contribution of the AI to clinical prediction domains. Table 4 and 5, in the Appendix, show the detailed extracted information from the 74 studies, regarding their objectives, design, speciality, sample size, population and settings, intervention and exposure, outcome measures, AI or machine learning model used, key findings, limitations, and conclusions. n addition to the eight domains identified, the analysis of the 74 studies showed that Oncology and Radiology come on top of the specialties where AI enhances clinical prediction. Table 3 and Fig. 4 show the medical specialties discussed in the 74 studies.

Table 1
The AI eight domains for clinical prediction.

SN	Domains	Functions
1	Diagnosis	Predicting the presence or absence of a disease or condition based on symptoms, clinical tests, and other patient data. Diagnostic predictions are essential for early detection and timely treatment.
2	Prognosis	Predicting the likely course and outcome of a disease or condition once it has been diagnosed. Prognostic predictions help in understanding the disease's progression, potential complications, and the likely response to treatment.
3	Risk Assessment	Predicting the likelihood of a patient developing a disease or condition in the future based on various risk factors such as genetics, lifestyle, environmental exposures, and existing health conditions.
4	Treatment Response	Predicting how a patient will respond to a particular treatment or therapy. This is especially important in personalised medicine, where the goal is to tailor treatments to individual patients for maximum effectiveness.
5	Disease Progression	Predicting how a disease will progress over time is crucial for chronic diseases such as diabetes, heart disease, and neurological disorders. It helps in planning long-term treatment strategies and anticipating future care needs.
6	Readmission Risks	Hospitals and healthcare providers use predictive models to identify patients who are at high risk of being readmitted after discharge. This helps in providing targeted interventions to reduce readmission rates.
7	Complication Risks	Predicting the risk of complications, both during and after medical procedures or treatments, is vital for informed consent and for planning to mitigate those risks.
8	Mortality Prediction	In critical care and palliative care settings, predicting the risk of mortality is important for decision-making regarding treatment intensity and end-of-life care planning.

Discussion

Domain one: diagnosis

Diagnosis is a vital aspect of medical care where the goal is to accurately predict the presence or absence of diseases or medical conditions [9,10]. AI algorithms, particularly those utilising machine learning and deep learning, can analyse complex and vast datasets more quickly and accurately than traditional methods. For instance, in medical imaging, AI can detect minor and complex patterns in X-rays, MRI, and CT scans that might be missed by the human eye. This capability is especially crucial in identifying early stages of diseases like cancer, where early intervention can substantially improve prognosis [4,97]. Moreover, AI-driven diagnostic tools are being integrated into primary care and telemedicine. These tools can assist healthcare professionals in making quick and accurate diagnoses by analysing symptoms reported by patients and comparing them with large medical datasets. This can make quality medical care more accessible, especially in remote or underserved areas [98]. AI in diagnostics also holds promise in personalised medicine, where it can help in identifying genetic markers and other specific factors that influence disease manifestation and progression [6,99].

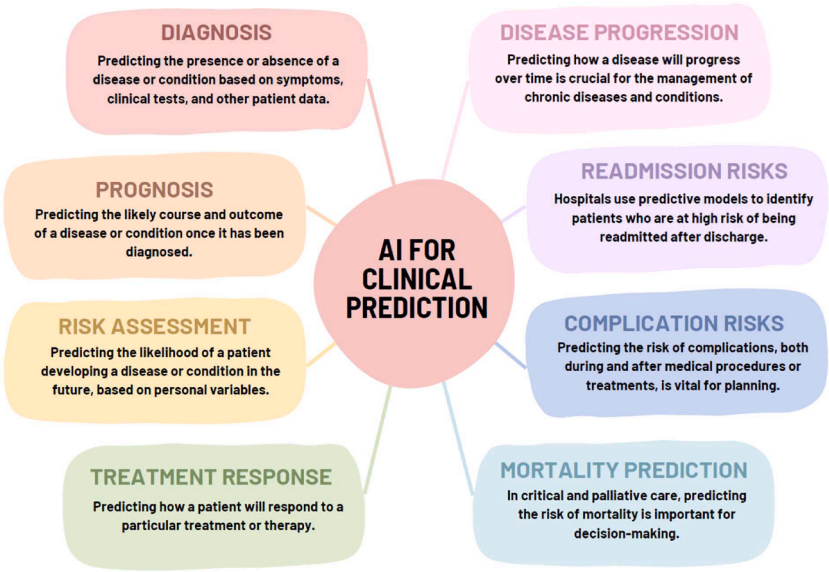

Fig. 2. AI supports eight domains of clinical prediction.

Domain two: prognosis

Prognosis involves predicting the future course and outcomes of a disease or condition after diagnosis [2,11]. In this domain, AI plays a transformative role. By leveraging large datasets and advanced algorithms, AI can identify patterns and correlations that may not be immediately apparent to human clinicians. This includes analysing progression trends of diseases, response rates to various treatments, and historical patient outcomes data. AI systems can also incorporate a wide range of variables, from genetic information to lifestyle factors, providing a more holistic and accurate prognosis [100]. For chronic and degenerative diseases like cancer, heart disease, and neurological disorders, AI-driven prognosis is particularly beneficial. It helps in understanding how these diseases are likely to progress over time, enabling healthcare providers to anticipate future complications and adjust treatment strategies accordingly. This proactive approach in managing chronic diseases can significantly improve patient quality of life and reduce long-term healthcare costs [101]. AI in prognosis also assists in personalised medicine. By understanding an individual patient's disease trajectory, treatments can be better tailored to their specific needs, improving efficacy and minimising adverse effects. This personalization is especially important in conditions with high variability in outcomes among patients [99].

Domain three: risk assessment

The risk assessment is a crucial domain in healthcare focusing on evaluating the probability of individuals developing a disease or condition in the future [12,13]. AI has become an indispensable tool in refining risk assessment. AI algorithms can process and analyse vast amounts of data, identifying risk factors and patterns that may not be immediately evident. For example, machine learning models can analyse genetic data alongside lifestyle and environmental factors to predict the risk of developing conditions like heart disease, diabetes, or certain types of cancer. This holistic approach provides a more comprehensive risk profile for each individual [4,9,102]. Moreover, AI-driven risk assessment is pivotal in public health initiatives. It aids in identifying populations at high risk for certain diseases, enabling targeted preventive measures and resource allocation. This approach is particularly effective in managing and preventing chronic diseases, which are a major challenge for global health systems [103]. AI in risk assessment also plays a role in personalised health recommendations. By understanding individual risk profiles, healthcare providers can offer tailored advice on lifestyle modifications, screening, and preventive measures. This personalised approach not only improves individual health outcomes but also contributes to more efficient and effective healthcare systems [104].

Domain four: treatment response

The treatment response is an essential facet of modern healthcare, focusing on predicting how patients will respond to specific treatments or therapies [14,15]. By leveraging advanced algorithms and large datasets, AI can analyse various factors that influence a patient's response to treatment. These factors include genetic makeup, biochemical parameters, lifestyle habits, and the presence of other health conditions. For instance, in oncology, AI models can predict how a cancer patient might respond to a particular chemotherapy regimen based on their genetic profile and the genetic characteristics of their tumour [105]. This predictive capability is not just limited to pharmacological treatments. AI systems are also being used to anticipate responses to surgical procedures, radiation therapy, and other medical interventions. This helps healthcare professionals in selecting the most appropriate treatment plan for each patient, maximising efficacy and reducing the risk of complications [106]. Furthermore, AI-driven treatment response prediction is instrumental in drug development. It enables researchers to identify which patient groups are most likely to benefit from new drugs, thereby facilitating more targeted and efficient clinical trials [107].

Domain five: disease progression

Disease Progression is a vital area in healthcare, especially for managing chronic conditions such as diabetes, heart disease, and neurological disorders [2,16]. AI has a significant impact in this area, offering advanced tools for monitoring and predicting disease progression. AI algorithms, particularly those using machine learning, can analyse large and complex datasets, including clinical records, patient histories, and biomedical data. This analysis helps in identifying patterns and markers that indicate the progression of a disease, sometimes even before clinical symptoms become apparent [108,109]. For instance, in the case of neurodegenerative diseases like Alzheimer's, AI can detect subtle changes in brain imaging or cognitive function tests that might predict the rate at which the disease will progress [110].

Table 2
Mapping the 74 studies to the eight clinical prediction domains.

SN	Study	Diagnosis	Prognosis	Risk Assessment	Treatment Response	Disease Progression	Readmission Risks	Complication Risks	Mortality Prediction
1	Wang et al., 2021 [23]	✓	✓						
2	Ma et al., 2022 [24]	✓			✓				
3	Huang et al., 2022 [25]	✓	✓						
4	Fremond et al., 2023 [26]	✓			✓				
5	Xu et al., 2021 [27]		✓						
6	Qin et al., 2021 [28]	✓	✓						
7	Salah et al., 2021 [29]				✓				
8	Shu et al., 2022 [30]	✓		✓					
9	Yue et al., 2022 [31]				✓				
10	Zhang et al., 2021 [32]	✓							
11	Groos et al., 2022 [33]	✓				✓			
12	Howell et al., 2021 [34]			✓	✓		✓		✓
13	Wen-Zhi et al., 2022 [35]	✓							
14	Cui et al., 2022 [36]		✓		✓				✓
15	Li et al., 2022 [37]		✓	✓					✓
16	Zhong et al., 2024 [38]		✓		✓	✓			
17	Liu et al., 2022 [39]			✓					
18	Feng et al., 2021 [40]	✓							✓
19	Yagi et al., 2022 [41]		✓			✓			
20	Min et al., 2021 [42]				✓				
21	El-Sappagh et al., 2021 [43]	✓							
22	Arabyarmohammadi et al., 2022 [44]			✓					✓
23	Liu et al., 2022 [45]		✓					✓	
24	Salari et al., 2023 [46]				✓				
25	Wen et al., 2023 [47]	✓	✓			✓			
26	Liu et al., 2021 [48]	✓							
27	Liu et al., 2023 [49]		✓						
28	Li et al., 2023 [50]		✓	✓	✓	✓			✓
29	Xia et al., 2023 [51]		✓			✓			✓
30	Vodencarevic et al., 2021 [52]		✓	✓			✓		
31	Li et al., 2022 [53]	✓	✓			✓			
32	Verma et al., 2022 [54]		✓	✓					
33	Hae et al., 2023 [55]			✓	✓				
34	Lee et al., 2022 [56]								✓

(continued on next page)

Table 2 (*continued*)

SN	Study	Diagnosis	Prognosis	Risk Assessment	Treatment Response	Disease Progression	Readmission Risks	Complication Risks	Mortality Prediction
35	Kong et al., 2023 [57]	✓							
36	Sundar et al., 2022 [58]		✓	✓	✓				
37	Sun et al., 2022 [59]	✓							
38	Zhang et al., 2022 [60]								✓
39	Fan et al., 2022 [61]	✓			✓				
40	Ou et al., 2022 [62]		✓					✓	
41	Luo Y et al., 2023 [63]		✓		✓				✓
42	Huang J et al., 2022 [64]	✓			✓			✓	
43	Yin P et al., 2023 [65]		✓			✓		✓	✓
44	Zhang Z et al., 2023 [66]								✓
45	Cheng M et al., 2023 [67]				✓				✓
46	Kao YT et al., 2023 [68]			✓					
47	Saux P et al., 2023 [69]		✓						
48	Li J et al., 2023 [70]		✓						
49	Faraone SV et al., 2022 [71]				✓				
50	Zhang K et al., 2023 [72]		✓						✓
51	Cai ZH et al., 2023 73]	✓							
52	Wang Y et al., 2023 74]			✓					
53	Bao Z et al., 2021 75]		✓		✓				
54	Li P et al., 2023 76]							✓	
55	Liu Y et al., 2023 77]		✓						
56	Xie N et al., 2023 78]		✓						
57	Chen X et al., 2023 79]			✓				✓	
58	Forrest LN et al., 2023 80]				✓				
59	Tan TH et al., 2021 81]	✓	✓						✓
60	Chandra RS et al., 2023 82]		✓		✓				
61	Jin Y et al., 2023 83]		✓		✓				
62	Su Z et al., 2023 84]			✓					
63	Liu Y et al., 2023 85]		✓					✓	✓
64	Wu Y et al., 2023 86]	✓							
65	Zhang K et al., 2022 87]	✓							
66	Tran QNN et al., 2023 88]		✓						✓
67	Zhang H et al., 2023 89]	✓	✓						
68	Ma X et al., 2023 90]	✓	✓						

(*continued on next page*)

Table 2 (*continued*)

SN	Study	Diagnosis	Prognosis	Risk Assessment	Treatment Response	Disease Progression	Readmission Risks	Complication Risks	Mortality Prediction
69	Li D et al., 2023 91]		✔						✔
70	Le Y et al., 2023 92]		✔			✔			✔
71	Zhang H et al., 2023 93]		✔						✔
72	Etter JF et al., 2023 94]						✔		
73	Lyu Z et al., 2023 95]		✔					✔	
74	Huang J et al., 2023 96]		✔						✔
Studies discussing each domain		24	38	14	22	9	3	9	20

✔ = Discussed domains in the 74 studies.

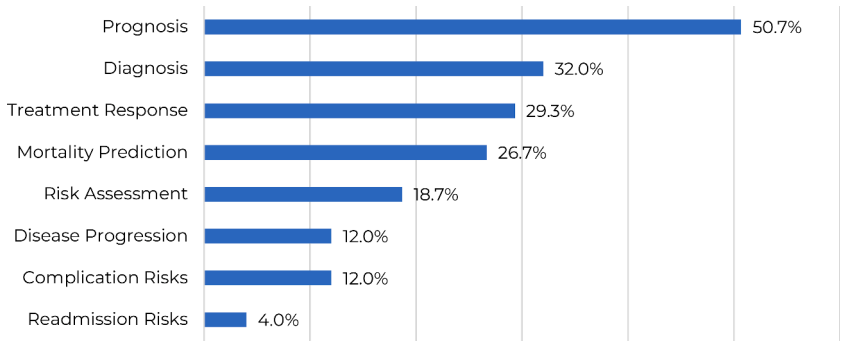

Fig. 3. AI contribution to clinical prediction domains, based on the 74 studies.

Similarly, in diabetes management, AI models can analyse blood sugar levels, lifestyle factors, and treatment responses to predict potential complications and guide adjustments in treatment plans [111,112]. AI's predictive capabilities in disease progression are not just limited to direct health outcomes. They also extend to predicting the impact of the disease on a patient's quality of life, daily functioning, and mental health. This comprehensive approach allows healthcare providers to offer more holistic and patient-centred care [113,114].

Domain six: readmission risks

Readmission Risks addresses a critical challenge in healthcare: identifying patients who are at high risk of being readmitted to the hospital shortly after discharge. This domain is particularly significant because high readmission rates are often indicators of suboptimal care and can lead to increased healthcare costs [17,18]. AI plays a pivotal role in assessing readmission risks. By analysing extensive datasets, including patient medical histories, treatment details, discharge conditions, and socio-demographic factors, AI algorithms can identify patterns and risk factors associated with higher chances of readmission. This information is crucial for healthcare providers to intervene proactively [115]. For example, AI can help in identifying patients who might need additional support post-discharge, such as those with chronic conditions like heart failure or diabetes, or elderly patients with multiple health issues. Accordingly, healthcare teams can implement targeted strategies such as arranging follow-up appointments, providing additional patient education, and ensuring proper medication management [116]. Moreover, AI-driven insights into readmission risks enable hospitals to allocate resources more efficiently, focusing on high-risk patients while optimising care for those with lower risks. This targeted approach not only improves patient care but also reduces unnecessary hospitalizations, thereby alleviating the financial strain on healthcare systems [117].

Domain seven: complication risks

Complication Risks is focused on predicting the likelihood of complications arising during or after medical procedures or treatments. This predictive capability is crucial for enhancing patient safety and improving care outcomes [19,20]. By analysing diverse data sets, including patient medical histories, the specifics of their current medical conditions, details of planned procedures, and even broader demographic data, AI algorithms can identify patterns and risk factors that might lead to complications. This could range from post-surgical infections to adverse reactions to medications or treatments [118,119]. For instance, in surgical settings, AI can evaluate a patient's risk of complications based on factors like age, underlying health conditions, and the nature of the surgery. This information is invaluable for surgeons and medical teams in planning and executing surgical interventions, allowing them to take preemptive measures to minimise risks [120]. Furthermore, AI-driven predictions are vital for informed consent processes. By providing more accurate information about potential risks, healthcare providers can ensure that patients are better informed about the procedures they are undergoing, leading to improved patient satisfaction and trust [121].

Domain eight: mortality prediction

Mortality Prediction is a sensitive yet crucial aspect of healthcare, particularly relevant in critical care and palliative care settings. This is essential for making informed decisions regarding treatment intensity and end-of-life care planning [21,22]. By analysing complex and comprehensive data sets, including patient medical histories, current health status, treatment responses, and even genetic information, AI

Table 3
Specialties where AI supports clinical prediction, based on the 74 studies.

Speciality	Number of studies
Oncology	32
Radiology	23
Neurology	11
Surgery	10
Cardiology	8
Pulmonology	5
Critical care medicine	4
Orthopaedics	4
Geriatrics	3
Psychiatry	3
Gerontology	3
Rheumatology	3
Gastroenterology	3
Urology	2
Infectious diseases	2
Pathology	2
Hepatology	2
Emergency medicine	2
Ophthalmology	2
Genetics	1
Psychology	1
Vascular medicine	1
Otolaryngology	1
Hospital management	1
Epidemiology	1
Public health	1
Addiction medicine	1
Diabetes	1
ADHD	1
Infectious disease	1
Dermatology	1
Bariatric surgery	1
Gynaecology	1
Trauma	1
Pain management	1
Endocrinology	1
Haematology	1
Neurosurgery	1
Clinical research	1
Paediatrics	1
Behavioural science	1

algorithms can identify patterns and indicators that may signify a higher risk of mortality. This predictive capability is particularly important in intensive care units, where rapid decision-making is often required [122]. AI's contribution to mortality prediction is not just about identifying those at highest risk; it also involves ensuring that patients receive appropriate levels of care. For instance, in cases where recovery is unlikely, such as patients with advanced stages of cancers, AI can help in recognising the need to shift from aggressive treatments to palliative care, focusing on the quality of life and comfort [123]. Moreover, AI-enabled mortality predictions are crucial for resource allocation in healthcare. Understanding which patients are most at risk helps in prioritising care and making critical decisions about the allocation of intensive care resources, especially in situations like pandemics or other health crises [124].

Technology landscape and innovations

The technology landscape associated with AI in clinical prediction spans various sophisticated techniques, notably Machine Learning (ML), Deep Learning (DL), and Natural Language Processing (NLP) [125]. Technological innovation in AI-supported clinical prediction, specifically through the application of Deep Learning Techniques and Large Language Models (LLMs), represents a significant leap forward in the medical field's capacity to diagnose diseases, predict outcomes, and tailor treatments to individual patients. Deep learning algorithms excel in analysing complex, large-scale datasets, including medical imaging,

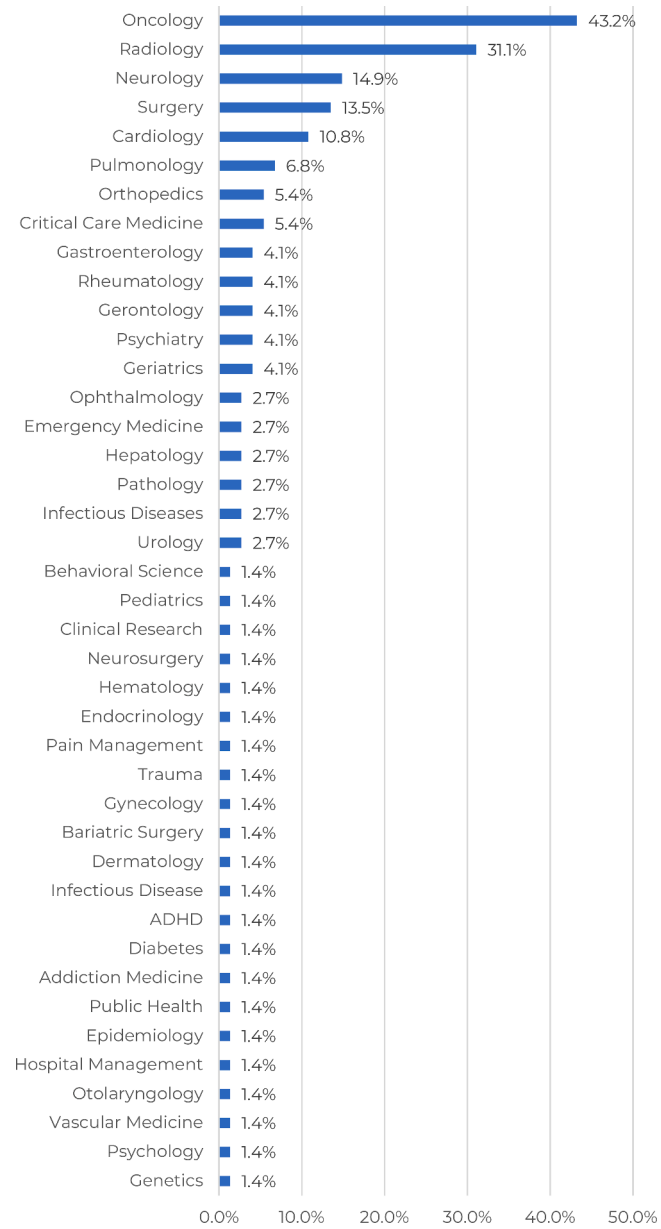

Fig. 4. Specialties where AI supports clinical prediction, based on the 74 studies.

genetic sequences, and electronic health records, uncovering patterns and anomalies that might be overlooked by traditional analytical methods [126]. This capability is crucial for early detection of diseases such as cancer, where timely intervention can dramatically improve prognosis. On the other hand, LLMs, such as GPT and BERT, have transformed the way medical professionals interpret vast amounts of unstructured text data, from clinical notes to medical research papers, enabling a deeper understanding of patient conditions and the medical landscape at large [127]. These advancements not only enhance diagnostic accuracy and prognostic predictions but also facilitate a more personalized approach to healthcare, aligning treatments with individual patient profiles for optimal outcomes. This synergy between AI technologies and medical expertise is paving the way for a future where healthcare is more efficient, effective, and patient-centred [6,9].

Conclusion and recommendations

AI has significantly advanced clinical prediction through its ability to

process complex datasets. In diagnostics, AI enhances early disease detection accuracy, facilitating timely interventions. It aids in prognosis by predicting disease progression and outcomes, supporting more effective treatments. AI's analysis of various factors improves risk assessment for disease development, essential in preventive medicine. In personalised medicine, AI is key in predicting treatment responses, customising care to individual needs. It's also beneficial in managing chronic diseases, anticipating progression for proactive care. AI tools identify patients with high readmission risks, supporting targeted interventions, and improve complication risk predictions to enhance patient safety. Furthermore, AI offers critical insights in mortality prediction.

Based on the findings of this review, the following eight recommendations are proposed to optimise the integration of AI in clinical prediction. (1) Enhance Data Quality and Accessibility: Ensure the collection of high-quality, diverse, and comprehensive healthcare data to train AI models effectively. Improved data sharing protocols should be established to facilitate access to diverse datasets while maintaining patient privacy and data security. (2) Promote Interdisciplinary Collaboration: Encourage collaboration between AI experts, healthcare professionals, and researchers. This collaboration will foster the development of AI tools that are clinically relevant and user-friendly for healthcare providers. (3) Focus on Ethical and Transparent AI Practices: Develop AI systems with ethical considerations and transparency. This includes addressing biases in AI algorithms, ensuring equitable healthcare delivery, and maintaining transparency in AI decision-making processes. (4) Invest in AI Education and Training: Provide education and training for healthcare professionals on AI tools and their applications in clinical prediction. This will enhance their ability to interpret AI-generated predictions and integrate them into clinical practice. (5) Expand Clinical Trials and Research: Conduct extensive clinical trials to validate the efficacy and safety of AI applications in clinical prediction. Further research should also focus on exploring the potential of AI in unexplored areas of healthcare. (6) Regulatory Oversight and Guidelines: Develop and implement regulatory frameworks and guidelines to govern the use of AI in clinical prediction. This will ensure that AI tools meet quality and safety standards before their clinical implementation. (7) Patient Engagement and Consent: Involve patients in the AI integration process, ensuring they are informed about how AI is used in their care. Patient consent should be sought, especially in cases where AI influences critical healthcare decisions. (8) Continuous Monitoring and Improvement: Regularly monitor the performance of AI systems in clinical settings and make iterative improvements based on feedback from healthcare providers and patients.

Declaration on the use of AI in the writing process

The authors of this manuscript declare that in the writing process of this work, no generative artificial intelligence (AI) or AI-assisted technologies were used to generate content, ideas, or theories. We utilized AI solely for the purpose of enhancing readability and refining language. This use was under strict human oversight and control. After the application of AI technologies, the authors carefully reviewed and edited the manuscript to ensure its accuracy and coherence. The authors understand the potential of AI to generate content that may sound authoritative yet might be incorrect, incomplete, or biased. Considering this, the authors ensured that the manuscript was thoroughly revised by human eyes and judgement. In line with Elsevier's Authorship Policy, the authors confirm that no AI or AI-assisted technologies have been listed as an author or co-author of this manuscript. The authors fully comprehend that authorship comes with responsibilities and tasks that can only be attributed to and performed by humans, and authors have adhered to these guidelines in the preparation of this manuscript.

CRediT authorship contribution statement

Mohamed Khalifa: Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Mona Albadawy:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare no conflicting interests to declare regarding the publication of this manuscript.

Supplementary materials

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